INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

WELDING MACHINE, ARC, DC, GENERATOR, CONSTANT CURRENT, CONSTANT VOLTAGE, INTERNAL COMBUSTION ENGINE DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers internal combustion engine driven arc welding machines of the generator design, with direct current, constant current and constant voltage output.

1.2 Classification. The welding machine shall be of the following types and sizes. The types and sizes to be furnished shall be as specified (see 6.2.1).

Type I - Air-cooled gasoline engine driven.

Size 200 - 200 amperes

Type II - Liquid-cooled gasoline engine driven.

Size 200 - 200 amperes
Size 300 - 300 amperes
Size 400 - 400 amperes
Size 500 - 500 amperes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type III - Liquid-cooled diesel engine driven.

Size 200 - 200 amperes
Size 250 - 250 amperes
Size 300 - 300 amperes
Size 400 - 400 amperes
Size 500 - 500 amperes

Type IV - Air-cooled diesel engine driven.

Size 200 - 200 amperes

1.3 Part number. Welding machines covered by this specification are identified by the following specification-based part numbering system:

M80165-#-###

Size (see 1.2).

Type (see 1.2).

Specification identifier.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. The issue of these documents shall be the latest revision of the document on the date of the solicitation and may include title changes.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

FEDERAL

W-C-596/40 - Connector, Receptacle, Electrical, General Purpose, Duplex, Hospital Grade, Grounding, 2 Pole, 3 Wire, 20 Amperes, 125 volts, 50/60 Hertz.

W-C-596/41 - Connector, Receptacle, Electrical, General Purpose, Single, Hospital Care Grounding, 2 Pole, 3 Wire, 20 Amperes, 125 volts, 50/60 Hertz.

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STANDARDS

MILITARY

MS-52110 - Trailer, Flatbed, 4 ton, 4 wheel, Military Design - M795.

MS-52146 - Chassis Trailer 1/4 ton, 2 wheel Military Design, M569A1.

MS-52152 - Chassis Trailer 3/4 ton, 2 wheel Military Design, M116A2.

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents and publications. The following other Government document forms a part of this specification to the extent specified herein. The issues shall be those in effect on the date of the solicitation and may include title changes.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of these documents shall be the latest revision of the document on the date of the solicitation and may include title changes.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA EW 1 - Electrical Arc-Welding Apparatus.

ANSI/NFPA 70 - National Electrical Code.

IEC 501 - Safety Requirements for ARC Welding Equipment Plugs, Socket - Outlets and Couplers for Welding Cables.

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(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

SOCIETY OF AUTOMOTIVE ENGINEERS INC.

SAE J684B - Trailer Couplings and Hitches, Automotive Type

SAE J847 - Full Trailer Tow Bar Type.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 485 Lexington Ave., New York, NY 10017)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The welding machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The welding machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Design and construction measurement systems. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the welding machine. In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials provided such materials will not jeopardize the welding machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition

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and physical properties as the materials originally selected for use on the welding machine.

3.2.3 Energy efficiency. The welding machine and its components that directly consume energy in normal operation shall be economically designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for the point of operation and all other parts of the welding machine that present safety hazards. The safety devices shall not interfere with the operation of the welding machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All welding machine parts, components, mechanisms, and assemblies furnished on the welding machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 that are applicable to the welding machine itself.

3.2.5 Mercury restriction. The welding machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the welding machine.

3.2.7 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.2.8 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.9 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.2.10 Maintainability. All parts and components requiring routine maintenance and adjustment shall be designed to be readily accessible and adjustable with simple tools.

3.2.11 Vibration. The engine and generator assembly shall be designed to operate with vibration characteristics below the level which will adversely affect the operation of the machine.

3.2.12 Stability. The machine shall be designed to operate satisfactorily when inclined at an angle of 15 degrees in any direction from the normal operating position.

3.2.13 Operating speed. The full-load operating speed of the machine shall be not greater than 3600 revolutions per minute (rpm).

3.2.14 Portability. The welding machines shall be portable, and shall be base mounted. Running gear, if required (see 6.2.11), shall be as specified by the

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procuring activity. The welding machine shall be provided with an identified lifting point above the center of mass for handling by crane or hoist and shall be capable of supporting 2-1/2 times the weight of the basic machine. The center of gravity of the entire unit shall be as low as is consistent with economical design.

3.3 Construction. The welding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the welding machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts, either US or SI, used on the welding machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.4 Components. The welding machines shall consist of not less than the following components: Engine, fuel system, generator, generator exciter, wiring, controls, auxiliary power supply, and enclosure.

3.4.1 Engine. The engines for the welding machines covered by this specification shall be standard commercial internal combustion engines of current design and development. The engine shall develop horsepower to operate the welding generator and accessories at maximum rated capacity. The engine for the type I machine shall be a 4-stroke cycle forced air-cooled gasoline engine which will operate satisfactorily on a lead free grade of not less than 85-octane gasoline. The engine for a type II machine shall be a 4-stroke cycle, liquid-cooled, gasoline engine which will operate satisfactorily on a lead free grade of less than 85-octane gasoline. The engine for a type III machines shall be 2- or 4-stroke cycle, liquid-cooled, diesel engine which will operate satisfactorily on #2 diesel fuel. The engine for the type IV machine shall be a 4-stroke cycle, air-cooled diesel engine which will operate satisfactorily on #2 diesel fuel.

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3.4.1.1 Starting system. The type I machine shall be manually started by a crank, automatic recoil-cord pull, or hand-wind pull. When specified (see 6.2.1), a 12-volt electric starting system shall be furnished on type I machines. The type I and II machines shall have a manual-choke control to aid in starting the engine. The type II, III and IV machines shall be furnished with a 12-volt electric starting system and a dry-charge or maintenance-free battery with adequate capacity to crank the engine under all conditions specified herein.

3.4.1.2 Engine lubrication system. The engine lubrication system shall provide lubrication for all moving engine parts for continuous operation. The type I machine shall have a splash-or pressure-type system. The type II and III machines shall have a full pressure-type system. All type machines shall have oil-drain provisions in the machine base for draining the engine crankcase. A bayonet gauge or dip stick that is calibrated for determining crankcase oil level shall be provided. When specified (see 6.2.1), all machines with an oil-pressure system shall be equipped with a low-oil pressure shut-down device to stop the engine in the event of inadequate oil pressure.

3.4.1.3 Ignition system. The type I machine with manual start shall have magneto ignition. The type I machine shall be equipped with 12-volt battery or magneto-type ignition when the electric starting system in 3.4.1.1 is specified. The type II, III and IV machine shall have 12-volt battery ignition.

3.4.1.4 Battery charging system. The type II, III and IV machines shall have a battery charging generator or alternator, and regulator of sufficient size and capacity to keep the battery charged for starting the engine in accordance with standard commercial practice. A battery charging system shall be provided on type I machines when the electric starting system in 3.4.1.1 is selected.

3.4.1.5 Governor. The machines shall have a constant speed mechanical or electronic type governor which shall maintain the speed of the engine within 5 percent of throttle setting at rated operating speed. The governor shall be belt or gear driven. Belt driven governor shall be protected for belt breakage overspeed.

3.4.1.6 Starting aid. When specified (see 6.2.1), the type III machine shall have a volatile-fluid type or glow-plug, cold-weather starting device. The volatile type shall be designed to inject a measured amount of starting fluid through a connection into the air intake manifold.

3.4.1.7 Air cleaner. The engines shall have an air cleaner to filter all air for the air intake system. The air cleaner shall be of the wet or dry type, designed for easy disassembly and servicing. When the air cleaner is mounted on the outside of the machine enclosure, the cleaner shall be removable for shipping and crating. A permanent maintenance label listing the air filter element number shall be affixed to the inside panel of the machine.

3.4.1.8 Oil filter. The type II, III and IV machines shall have an oil filter of the renewable cartridge type. The part number of the cartridge shall be permanently and legibly marked on the filter body, cap or on a maintenance label which is affixed to the inside panel of the machine.

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3.4.1.9 Cooling system. The type I and IV engines shall be air cooled through shrouding and means of passing air over the cylinder block and head. The engines for the type II and III machines shall have a liquid cooling system with thermostatic control. The radiator and cooling fan shall be of adequate size and capacity so that the machine can operate in ambient temperatures ranging from -25 degrees F through +100 degrees F. The outside of the radiator shall be protected from damage by a radiator guard. When specified (see 6.2.1), the machine shall have a high temperature shutdown device to stop the engine in the event of excessive temperature.

3.4.1.10 Automatic idling device. Automatic idling device shall be provided on the welding machine to reduce the speed of the engine to an economical idle when not welding. When operating the machine in the constant current mode and upon striking an arc, the idling device shall immediately allow the engine to speed up to its previous setting. The idling device shall have the means to keep the engine at a constant speed if so desired. The idling device shall have a time delay to allow the operator, when welding, to change electrodes before the engine can come to idling speed (see 6.2.1).

3.4.1.11 Muffler. The machines shall have a muffler on the engine exhaust system. If the exhaust system is the vertical or upright type, it shall have a suitable rain cap on the end of the exhaust pipe to prevent water from entering the exhaust pipe.

3.4.2 Fuel system.

3.4.2.1 Fuel tank. The tank shall have fuel capacity to operate the type I and IV machine 3 hours at rated-duty cycle, rated load; and type II and III machines 8 hours at rated-duty cycle, rated load. The filler cap shall be chain connected to the tank and a drain plug provided on type II, III and IV machines.

3.4.2.2 Fuel quantity gage. The type II and III machines shall have a gage to register the quantity of fuel in the tank at all times. The scale on the gage shall be marked in quarter increments, such as 0, 1/4, 1/2, 3/4, F.

3.4.2.3 Fuel pump. The type II, III and IV machines shall have a fuel pump to supply fuel to the engine when the fuel system is not a full-gravity feed system. The fuel pump shall be mechanical or electrical. All machines shall be equipped with a fuel feeding system capable of operating at full load for all operating conditions.

3.4.2.4 Fuel lines. Fuel lines shall be installed free of sharp bends and loops that might cause failure from fatigue, strain, or vapor lock. The fuel lines shall be bracketed and fastened when necessary to prevent vibration. A shut-off cock shall be installed in the outlet of the tank when the tank is located to provide gravity feed.

3.4.2.5 Fuel strainer and filter. Type I and II machines shall have a strainer or in-line fuel filter. The strainer for type II machines shall have a removable cover or bowl for cleaning and servicing. The cover or bowl gasket shall be made of fuel-resistant rubber. Type III and IV machines shall have means to filter the

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fuel before it enters the fuel injection pump. The filter shall be of the replaceable cartridge type.

3.4.3 Welding generator. The generator shall conform to the standards in NEMA EW-1. The rotor shaft shall be mounted in a double-sealed antifriction bearing. The rotor shall be close coupled to the engine through a metal coupling to maintain alignment with the engine flywheel.

3.4.3.1 Generator output terminals. The terminals shall be not less than 1/2-inch in diameter and shall be recessed, protected, or located to prevent accidental contact. The electrode and the work terminals shall be so identified. Separate terminals or other means shall be provided for changing from constant current to constant voltage operation.

3.4.3.2 Electrical insulation. The class of electrical insulation system used in the generator and exciter shall be not less than class B of NEMA EW-1.

3.4.4 Generator exciter. The exciter shall be intergral with the welding generator, or be separate and mounted on the same shaft as the main welding generator. The exciter shall have fast voltage build up, and be designed to prevent polarity reversal. The commutator, brushes, and holders shall be readily accessible for servicing and maintenance.

3.4.5 Wiring. The type and size of wire used shall have sufficient capacity, temperature and voltage rating for the intended use. Wiring subject to chafing or vibration shall be secured to prevent damage. The wire and terminals shall be color or number coded. A wiring diagram of the generator, the exciter and engine shall be provided and fastened in a convenient location to the inside surface of the machine enclosure. The diagram shall have a transparent covering or protective coating.

3.4.6 Machine mounting. Unless otherwise specified, the welding machine shall be furnished with a standard metal base, arranged and drilled for bolting the machine to the floor or deck. When required, the machine shall be mounted on skids or mounted on a trailer for portability in a shop or work area as specified (see 6.2.1).

3.4.6.1 Trailer. When specified (see 6.2.1), the machine shall be mounted on a two-wheel or four-wheel tandem trailer with tube or tubeless tires. The tires shall be standard pneumatic commercial type having a ply rating and size recommended in the tire and rim association year book for the gross weight of the unit. The trailer shall be designed in such a manner that, when fully loaded, it may be towed at speeds up to 50 miles per hour on improved roads and up to 10 mph on rough terrain. When being towed at any speed up to 50 mph, the trailer shall track-true with minimum side sway or hunting and shall support the complete welding machine and accessories. Road clearance to the trailer's lowest member shall not be less than 8 inches. The trailer shall have resilient mounting (springs, torsion bars, or similar devices) to minimize transmission of road shock to the welding machine. When specified (see 6.2.1), a spare tire and wheel shall be provided and mounted in an accessible area on the trailer. When specified (see 6.2.1) the trailer must conform to one of the following detail specification sheets (MS) dependent upon the gross weight of the trailer, welder, accessories, miscellaneous tools, and other supplies which may be carried:

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MS-52146(AT), Chassis Trailer 1/4 Ton, 2 Wheel Military Design, M569A1
MS-52152(AT), Chassis Trailer 3/4 Ton, 2 Wheel Military Design, M116A2
MS-52110, Trailer, Flatbed, 4 Ton, 4 Wheel, Military Design, M-795

3.4.6.1.1 Provisions for accessories. Space shall be provided on over the road trailer for firmly mounting two gas cylinders and storage of all hoses and welding accessories. Gas cylinder mounting assembly(s) shall be provided to accommodate one gas cylinder up to 9-1/4 diameter and one gas cylinder up to 12-7/8-inch diameter. The mounting assembly(s) shall be positioned so as not to interfere with the control panel, hinder daily servicing, or add to the design volumn when the gas cylinders are installed. The installed gas cylinders shall not affect the fore and aft balance of tandem axle units.

3.4.6.1.2 Drawbar. The drawbar tongue shall be approximately 48 inches long and shall have a lunette-type ring fabricated on 1-inch round steel. The ring shall have an inside nominal diameter of 3 inches.

3.4.6.1.3 Rear parking stabilizer. The trailer shall be equipped with a rear mounted parking stabilizer or parking stand, which is adjustable for variations in terrain, and capable of supporting a 600-pound load (see 4.7.5).

3.4.6.1.4 Lights reflectors and chains. Vehicle reflectors, lights and safety chains shall be installed on the trailer in accordance with the manufacturer's standard practice but shall comply with SAE J847 and SAE J684B.

3.4.6.1.5 Parking brake. The trailer shall be equipped with a parking brake sufficient to hold the fully loaded trailer on a 15-degree incline.

3.4.6.1.6 Landing gear. Trailer shall be equipped with a retractable or adjustable type landing gear with full swiveling bottom wheel. A height adjustment of not less than 4 inches above level to 4 inches below position shall be provided.

3.4.7 Controls and instruments. The controls and instruments shall be grouped together on a panel, and shall be legibly marked for identification and position. Type II and III machines shall have a temperature gage and oil-pressure gage or warning lights. Machines with an electric starter system shall have an ammeter to indicate a charging rate of the generator or alternator. A voltmeter and an ammeter shall be provided to indicate the output voltage and welding current. The instruments shall be accurate with 3 percent of full-scale reading. The instruments shall be flush-mounted panel type, designed for protection against dust and moisture. Controls shall be provided to set the welding current in five steps or greater with fine adjustments within each welding range of the machine. The controls shall be designed for operation with a glove or mittened hand. Engine speed controls shall include a manual or automatic throttle to bring engine speed to a slow idle. An off-on switch for type I machines shall be provided.

3.4.8 Auxiliary power supply/generator. When specified the machine shall include an auxiliary power supply or generator which shall supply 120 + or -12 volts, 60 + or - 1 Hertz, 30 amperes (minimum), single-phase electric power to two grounding type, 20-ampere, 125-volt, single receptacles conforming to W-C-

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596/41 or one grounding type, 20-ampere, 125-volt, duplex receptacle conforming to W-C-596/40 which is wired as two independent receptacles. The circuit shall be provided with overload protection.

3.4.9 Enclosure. The machine shall have a weatherproofed enclosure, or be weatherproof. The type II and III machines shall have hinged or removable panels for access and operation. Where operating design requires, the enclosure shall be such that internal parts are protected from the weather even when the enclosure is opened in the operating position. The sheet metal of the enclosure shall be not less than 16 gage (0.0568 in.) in thickness.

3.4.10 Solid-state components. Solid-state design shall be used throughout for electronic components.

3.5 Performance. The welding machine shall be capable of providing direct-current (DC) power for welding; provide portability in shops, field maintenance and on construction site; and to provide auxiliary electric power as desired in connection with the welding operation.

3.5.1 Constant current operation. The welding machines shall be capable of operating as a constant-current welding machine for manual welding, with covered electrodes of all sizes within the machine capacity. The welding power range shall be in accordance with requirements of NEMA EW 1, class I machines. The welding machines shall be capable of supplying power for air arc gouging at welding power equal to the machine's rated capacity.

3.5.2 Constant voltage operation. The welding machines shall be capable of operating as constant-voltage welding machines, with the use of gas metal-arc welding sets and other automatic and semi-automatic wire feed consumable electrode welding sets. When adjusted for constant-voltage operation, the welding machines shall produce welding power with a flat volt-ampere curve.

3.5.3 Open circuit voltage. The open circuit voltage shall be in accordance with NEMA EW 1 for constant current operation. For constant voltage operation, the open-circuit voltage range shall be 25 volts or less to 50 volts or more. The maximum open-circuit voltage for either mode of operation shall not exceed 100 volts.

3.5.4 Operating temperature. The welding machines shall conform to the operating temperatures for the class of insulation used as specified in NEMA EW 1.

3.6 Standard equipment. Unless otherwise specified, all standard equipment normally furnished by the manufacturer shall be furnished. Additional equipment shall be furnished as specified and fully described (see 6.2.1).

3.6.1 Accessories. The following accessories shall be furnished:

3.6.2 One set of 4/0 flexible, concentric standard, copper welding cables for use on all size machines shall be furnished. The cables shall be fitted with an electrode holder, a ground clamp, and two cable lugs to mate with the terminals on the machine. Male and female plug type connections for cable ends conforming to IEC-501 shall be furnished. Cable insulation must be tested to assure adequacy for handling maximum open circuit voltage; water, ozone, oil, and solvent

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resistance; and voltage drop not exceeding 3.41 volts DC per 100 feet. When required, cables shall be tested for resistance and cross sectional area (see 4.7.2.2). Welding cables shall conform to NFPA-70.

3.6.2.1 When specified (see 6.2.1), provisions on the machine for storage of the welding cables, remote control unit, and tools shall be provided.

3.6.2.2 When specified (see 6.2.1), a stud welding gun and connectors compatible with the welder (s) concerned shall be provided.

3.7 Optional equipment. Optional equipment shall be furnished as specified (see 6.2.1).

3.8 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.9 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.9.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each welding machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.9.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each welding machine and contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Engine make, model
Constant current characteristics
Constant voltage characteristics
Auxiliary electric power KVA, voltage, AC or DC
Date of manufacture, year and month
Contract or Order Number
National Stock Number, or Plant Equipment Code

3.9.3 Hour meter. Unless otherwise specified (see 6.2.1), the welding machine shall be equipped with an hour meter. The hour meter shall be installed to

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display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the welding machine.

3.10 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.11 Workmanship. Workmanship of the welding machine shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.7).

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in an indoor facility with ambient conditions of 55°F to 100°F and 10 percent to 95 percent relative humidity.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the

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examination in 4.6 and all tests in 4.7. The machine shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. The welding machine shall be examined to determine compliance with all requirements of this specification including compliance with all applicable documents cited in section 2.

4.7 Tests.

4.7.1 Operational test. The welding machine shall be operated at no load for not less than 1 hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance tests. The leads of the welding machine shall be connected to a resistance load bank that is rated in excess of the maximum potential of the generator. The welding machine shall be operated through successive periods of 10 minutes each in increments of 25, 50, 75, and 100 percent of rated output. The preceding test shall be applied to both modes of performance as specified in 3.5.1 and 3.5.2. The operating temperatures of the machine shall be recorded after the machine is thermally stable to determine compliance to 3.5.4. The open circuit voltage shall be in compliance to 3.5.3. At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following tests (see 6.2.1). All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.7.2.1 Rain test. The machine shall be subjected to a weatherproof test by spraying fresh water at a rate of 4 inches $\pm$ 1 inch per hour at an angle of 30 to 45 degrees from vertical. The test shall be performed with water directed against all four sides and the roof of the machine simultaneously and continuously for not less than 1 hour. Immediately following the water spray, the machine shall be examined for water penetration and damage and the machine shall start and be operated to meet the performance requirements specified herein.

4.7.2.2 Lead cable test. When specified (see 6.2.1), lead cables shall be tested to determine conformance as indicated below. A Certificate of Conformance from an independent laboratory acceptable to the Government may be accepted in lieu of testing.

a. Direct current resistance. Tested by method for direct-current resistance of conductors. Resistance shall not exceed the parameters for portable arc welding copper cables.

4.7.2.3 Stud welding bend test. When a stud welding gun is required with the machine the gun shall be tested to demonstrate stud welding capabilities. Two studs or each diameter (1/8, 1/4, 3/8, or 7/16 inch), depending on welder involved

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(see 3.6.2.2) from cold drawn bar stock of 55,000 pounds per square inch (psi) minimum, shall be stud welded to a base member of thickness equal to 1/2-stud diameter and two studs each of applicable diameter shall be stud welded to a base member with thickness equal the stud diameter. Each welded stud shall be bent (by striking with a hammer) to an angle of 30 degrees from their original axis. Any cracking or breaking in the weld shall be cause for rejection.

4.7.3 Parking brake test. The fully loaded trailer shall be placed on a 15-degree incline and the parking brake applied. The engine of the welding machine shall be started and run at normal operating speed. There shall be no movement of the trailer down the incline under the above conditions.

4.7.4 Trailer road test. When a over-the-road type trailer is required it shall be tested for roadability. The fully loaded trailer shall be towed for 10 miles at approximately 50 mph on improved highways and 10 miles at approximately 10 mph on rough terrain. During this test, the trailer shall track steady, with miminum sidesway, and cause no difficulty in handling the towing vehicle. At the end of this test, the welding machine shall be examined for damage due to road shock. Failure to track steady, difficulty in handling the towing vehicle, or damage due to road shock shall be cause for rejection.

4.7.5 Trailer static stability test. When a trailer is required, it shall be tested for stability. The trailer shall be fully loaded with welder and all accessories identified in 3.4.6.1.1, and parked on level ground. The trailer shall be disconnected from the towing vehicle. Both two-wheel and four-wheel (tandem axle) trailers shall be leveled by adjustment of the landing gear. The rear parking stabilizer shall be extended. A 600-pound downward force shall be applied to the frame of the trailer at the four corners and the center of each side consecutively (left, right, front, and rear). With each application, the trailer shall remain stable, with no tendency to tip over during this test. The trailer shall fail this test if any component part is broken, permanently deformed, or if the trailer upsets (see 3.4.6.1.6). This test is in addition to those required by the applicable MS sheet covering the trailer referenced in 3 above.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packaging, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended to provide a DC power supply for welding where utility electric power is not available, where portability is desired in shops, field maintenance, and construction, and to provide auxiliary electric power as desired in connection with the welding operation.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. First article, if required (see 3.1).

d. If the welding machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. Running gear, if required (see 3.2.14).

f. Painting, if different (see 3.3.5).

g. Electric starting system for type I, if required (see 3.4.1.1).

h. Low oil pressure safety shut down, if required (see 3.4.1.2).

i. Starting aid for type III, if required (see 3.4.1.6).

j. High temperature safety shut down, if required (see 3.4.1.9).

k. Automatic idling device, if required (see 3.4.1.10).

l. Machine mounting, if different (see 3.4.6).

m. Over the road trailer mounting if required (see 3.4.6.1).

n. Detailed specification sheets (MS), if required (see 3.4.6.1)

o. Spare tire and wheel, if required (see 3.4.6.1).

p. Electrical system, if different (see 3.4.8).

q. Auxiliary power supply KVA capacity, voltage, and frequency, as required (see 3.4.8).

r. Standard equipment, if different (see 3.6).

s. Specify if provisions for storage of cable and accessories on the machine are required (see 3.6.2.1).

t. Stud welding gun when required (see 3.6.2.2).

u. Specify and fully describe optional equipment, if required see (3.7).

v. Specify applicable repair parts, if required (see 3.8).

w. Lubrication chart or plate, if different (see 3.9.1).

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x. Nameplate, if different (see 3.9.2).

y. Hour meter, if different (see 3.9.3).

z. First article inspection, if different (see 4.4).

aa. Quality conformance inspection, if different (see 4.5).

bb. Lead cable test, when specified (see 4.7.2.2).

cc. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level required (see 5.1).

6.2.2 Consideration of data requirements. The requirements for technical data should be considered when this specification is applied on a contract. The Data Item Descriptions (DIDs), DD Form 1664, for required data should be listed on a Contract Data Requirements List (DD Form 1423). The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements List, (AMSDL) should be researched for the applicable DD Form 1664 and to ensure that only current approved DIDs are cited on the DD Form 1423.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Subject term (keyword) listing.

Engine
Fuel System
Generator, welding
Trailer

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Custodians:
Army-AL
Navy-SH
Air Force - 99

Review activities:
Army - AL
Navy - YD, AS
Air Force - 84
DLA - GS

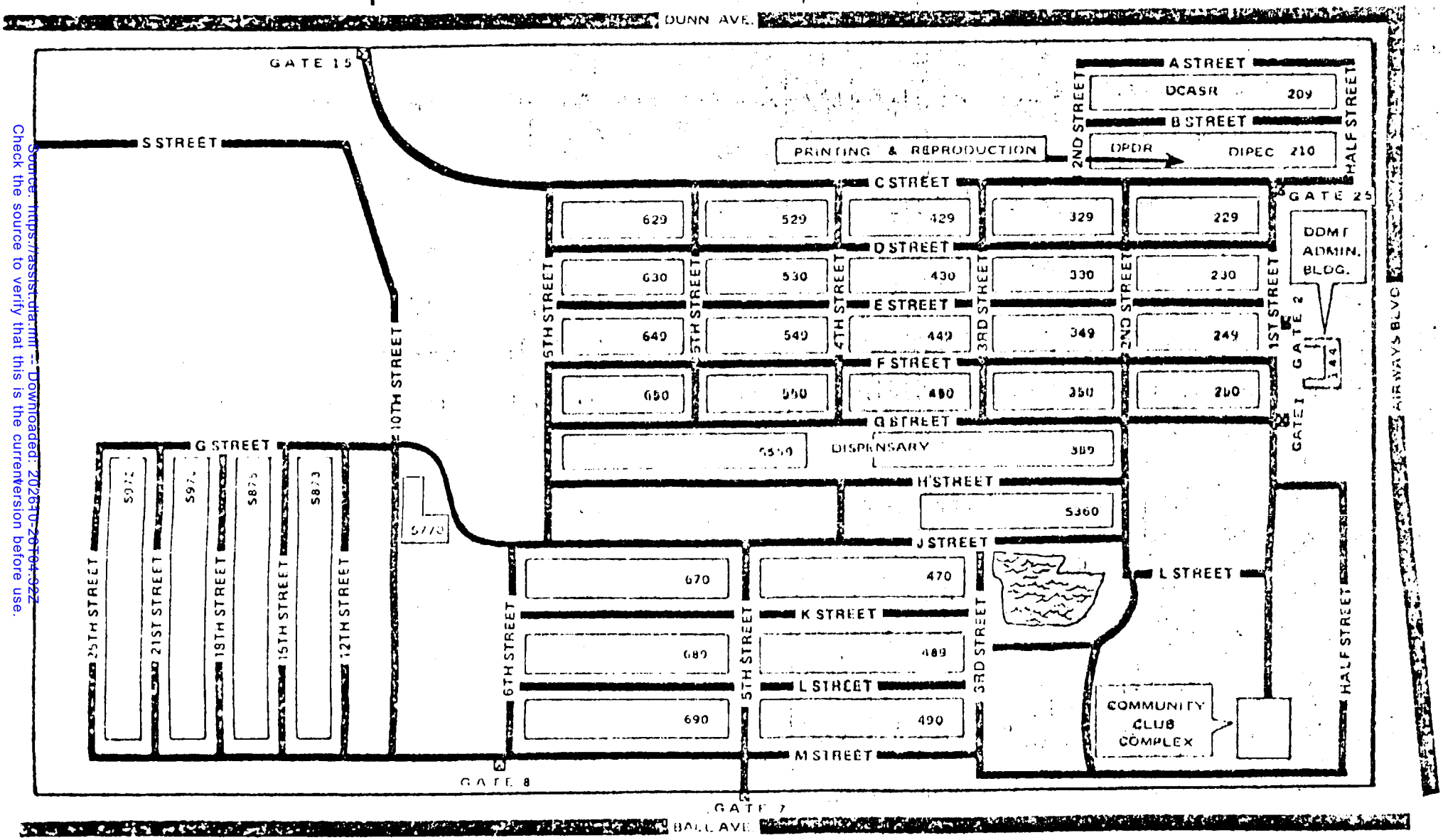
User activities:
Army - MV
Navy - MC, YD

Preparing activity:
DLA - IP

Project (3431-0178)

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Defense Depot Memphis Map



Speed Limit: 20 mph Except Where Posted

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80165B	Welding Machine, Arc, DD, Generator, Constant Current, Constant Voltage, Internal Combustion Engine Driven

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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